The power of least squares in the worst-case and learning setting

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Mathematik!
TU Chemnitz

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Applied functional analysis		Applied analysis
		
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Least squares method

Given: samples $(x^i, y_i) \in [0,1] \times \mathbb{R}$ for $i = 1, \ldots, n$

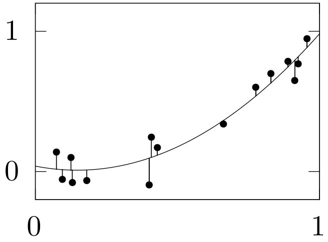
Goal: fit a function through the samples

Idea: we make the ansatz of a quadratic polynomial

$$g(x) = ax^2 + bx + c$$

and minimize

$$\sum_{i=1}^{n} |g(x^i) - y_i|^2$$



Least squares method

Given: samples $(\boldsymbol{x}^i, y_i) \in D \times \mathbb{C}$ for $i = 1, \ldots, n$ and $\emptyset \neq D \subset \mathbb{R}^d$

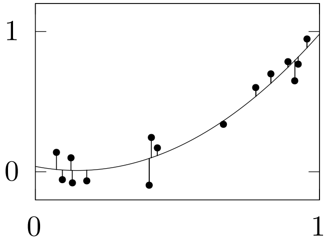
Goal: fit a function through the samples

Idea: for a set of functions $\{\eta_1, \ldots, \eta_{m-1}\} \subset \mathbb{C}^D$, we make the ansatz

$$g(\boldsymbol{x}) = \sum_{k=1}^{m-1} c_k \eta_k(\boldsymbol{x})$$

and for weights $\omega_i \geq 0$, we minimize

$$\sum_{i=1}^{n} \omega_i |g(x^i) - y_i|^2$$



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Implementation

- assume $n \geq m$ (oversampling)
- ansatz $g(\boldsymbol{x}) = \sum_{k=1}^{m-1} c_k \eta_k(\boldsymbol{x})$ with $\boldsymbol{c} = (c_1, \ldots, c_{m-1})^\mathsf{T}$ solving

$$\left\| \underbrace{\begin{pmatrix} \eta_1(\boldsymbol{x}^1) & \cdots & \eta_{m-1}(\boldsymbol{x}^1) \\ \vdots & \ddots & \vdots \\ \eta_1(\boldsymbol{x}^n) & \cdots & \eta_{m-1}(\boldsymbol{x}^n) \end{pmatrix}}_{=:\boldsymbol{L}} \begin{pmatrix} c_1 \\ \vdots \\ c_{m-1} \end{pmatrix} - \underbrace{\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}}_{=:\boldsymbol{y}} \right\|_{\boldsymbol{W}}^2 \to \min$$

  for a weight matrix $\boldsymbol{W} = \mathrm{diag}(\omega_1, \ldots, \omega_n)$
- the solution is given by

$$\boldsymbol{c} = (\boldsymbol{L}^* \boldsymbol{W} \boldsymbol{L})^{-1} \boldsymbol{L}^* \boldsymbol{W} \boldsymbol{y} \quad \text{and} \quad (S_{\boldsymbol{X}} \boldsymbol{y})(\boldsymbol{x}) = \sum_{k=1}^{m-1} c_k \eta_k(\boldsymbol{x})$$

Stability

Lemma

Let $\eta_1, \dots, \eta_{m-1}$ be an ONS of a function space $V \subset L_2$, $\boldsymbol{X}$ be points, and $\omega_1, \dots, \omega_M$ weights. Then T.f.a.e.

1. the system matrix is well-conditioned:
$$A \le \sigma^2_{\min}(\boldsymbol{W}^{1/2}\boldsymbol{L}) \le \sigma^2_{\max}(\boldsymbol{W}^{1/2}\boldsymbol{L}) \le B\,,$$
2. we have an **L_2-MZ inequality** for V, i.e.,
$$A\|g\|^2_{L_2} \le \sum_{i=1}^{M} \omega_i |g(\boldsymbol{x}^i)|^2 \le B\|g\|^2_{L_2} \quad \text{for all} \quad g \in V\,.$$

Stability

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MZ inequalities are widly available: [Mhaskar, Narcowich, Ward '01], [Nuyens, Cools '06], [Keiner, Kunis, Potts '07], [Filbir, Mhaskar '11], [Müller-Gronbach, Novak, Ritter '12], [Kämmerer, Potts, Volkmer '15], [Temlyakov '18], [Trefethen '19], [Gröchenig '20], [Filbir, Hielscher, Jahn, T. Ullrich to appear], ...

Stability

Theorem
[Nagel, Schäfer, T. Ullrich 2021][Moeller, T. Ullrich 2021][B 2022]

Let
- $\eta_1, \dots, \eta_{m-1}$ be an ONS of a function space $V \subset L_2(D, \nu)$,
- $n \in \mathbb{N}$, $t > 0$ be s.t. $6m(\log m + t) \le n$
- $\boldsymbol{X} = \{\boldsymbol{x}^1, \dots, \boldsymbol{x}^n\}$ be points drawn i.i.d. w.r.t. the density

$$\varrho(\boldsymbol{x}) = \frac{1}{m-1} \sum_{k=1}^{m-1} |\eta_k(\boldsymbol{x})|^2$$

- $\omega_i = 1/\varrho(\boldsymbol{x}^i)$.

Then we have an **L_2-MZ inequality** for V with constants $A = 1/2$ and $B = 3/2$.

Fast matrix-vector multiplication

domain D	basis $\eta_{\boldsymbol{k}}$	fast algorithm
$\mathbb{T}^d$	$\exp(2\pi\mathrm{i}\langle \boldsymbol{k}, \boldsymbol{x}\rangle)$	FFT, LFFT, NFFT
$[0, 1]$	$\cos(k \arccos x)$	FCT, NFCT
$\mathbb{S}^2$	$Y_{k,l}(\vartheta, \varphi)$	NFSFT
$\mathrm{SO}(3)$	$D_k^{ll'}(\boldsymbol{R})$	NSOFT

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Worst-case setting

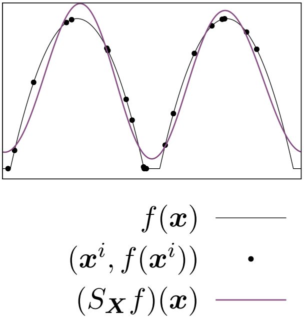
Given:

- $L_2 = L_2(D,\nu)$
- reproducing kernel Hilbert space $H(K) \hookrightarrow L_2$

Goal:

- find good points $\boldsymbol{X} = \{\boldsymbol{x}^1, \dots, \boldsymbol{x}^n\} \subset D$ and a sampling recovery operator $S_{\boldsymbol{X}} \colon H(K) \to L_2$ with small **worst-case error**

$$\sup_{\|f\|_{H(K)} \le 1} \|f - S_{\boldsymbol{X}} f\|_{L_2}$$

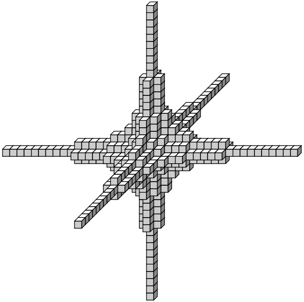


- $H^s_{\text{mix}}(\mathbb{T}^d) = \{f \in L_2 : \|f\|_{H^s_{\text{mix}}} < \infty\}$ $(s > 1/2)$
with

$$\|f\|^2_{H^s_{\text{mix}}} := \sum_{\boldsymbol{j}\in\{0,s\}^d} \|D^{(\boldsymbol{j})} f\|^2_{L_2}$$

- basis functions $\eta_{\boldsymbol{k}} = \exp(2\pi \mathrm{i}\langle \boldsymbol{k}, \cdot\rangle)$
- Kolmogorov width

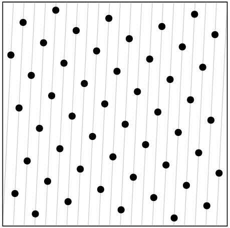
$$d_m(H^s_{\text{mix}}) := \inf_{\substack{\ell_1,\dots,\ell_m :\ H^s_{\text{mix}} \to \mathbb{C} \\ \varphi_1,\dots,\varphi_m \in L_2}} \sup_{\|f\|_{H^s_{\text{mix}}} \le 1} \Big\| f - \sum_{k=1}^{m} \ell_k(f) \varphi_k \Big\|_{L_2}$$
$$= m^{-s} (\log m)^{(d-1)s}$$



I^{hc}_R

Structured points: rank-1 lattice

- $\boldsymbol{X} = \Lambda(\boldsymbol{z}, n) = \left\{ \frac{i}{n} \boldsymbol{z} \bmod \mathbf{1}, i = 0, \ldots, n-1 \right\}$
- [Nuyens '07], [Kämmerer, Potts, Volkmer '15]:
  - given I, there exist algorithms that find $\boldsymbol{z}$ and n such that $\boldsymbol{L}^* \boldsymbol{W} \boldsymbol{L} = \boldsymbol{I}$
  - multiplication with $\boldsymbol{L}$ can be carried out with the LFFT in $\mathcal{O}(n \log n)$



$z = (1, 21)^{\mathsf{T}}, n = 55$

Theorem [Byrenheid, Kämmerer, T. Ullrich, Volkmer '17]

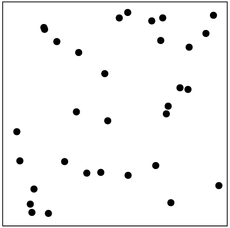
$$n^{-s/2} \lesssim \sup_{\|f\|_{H^s_{\text{mix}}} \le 1} \|f - S_{\boldsymbol{X}} f\|_{L_2} \lesssim n^{-s/2} (\log n)^{\frac{d-2}{2} s + \frac{d-1}{2}}$$

Random points

- $\boldsymbol{X} = \{\boldsymbol{x}^1, \ldots, \boldsymbol{x}^n\}$ drawn randomly i.i.d. w.r.t. Lebesgue measure such that

$$n \sim m \log m$$

- no fast matrix-vector multiplication due to lack of structure

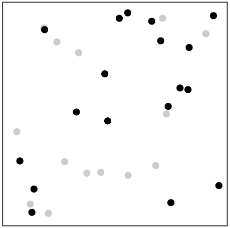


Theorem [M. Ullrich, Krieg '19]

$$n^{-s}(\log n)^{(d-1)s} \lesssim \sup_{\|f\|_{H^s_{\text{mix}}} \le 1} \|f - S_{\boldsymbol{X}} f\|_{L_2} \lesssim n^{-s}(\log n)^{(d-1)s+s}$$

Subsampled random points

- $\boldsymbol{X} = \{\boldsymbol{x}^1, \dots, \boldsymbol{x}^n\}$ drawn randomly i.i.d. w.r.t. Lebesgue measure such that $n \sim m \log m$
- there exist $\boldsymbol{X}' = \{\boldsymbol{x}^{i_1}, \dots, \boldsymbol{x}^{i_{n'}}\} \subset \boldsymbol{X}$ subsampled points suitable for reconstruction with
$$n' \in \mathcal{O}(m)$$
- no fast matrix-vector multiplication



Theorem [Nagel, Schäfer, T. Ullrich '21], [B., Schäfer, T. Ullrich '22]

$$n^{-s}(\log n)^{(d-1)s} \lesssim \sup_{\|f\|_{H(K)} \le 1} \|f - S_{\boldsymbol{X}} f\|_{L_2} \lesssim n^{-s}(\log n)^{(d-1)s+1/2}$$

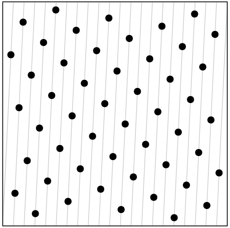
Theorem [Dolbeaut, Krieg, M. Ullrich '22]

$$\sup_{\|f\|_{H(K)} \le 1} \|f - S_{\boldsymbol{X}} f\|_{L_2} \sim n^{-s}(\log n)^{(d-1)s}$$

Subsampling quadratue points

1

$\boldsymbol{X}_{\mathsf{MZ}}$: good deterministic set of M points



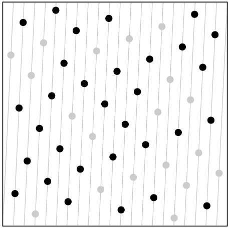
$\boldsymbol{X} = \{\boldsymbol{x}^1, \dots, \boldsymbol{x}^M\}$

$\boldsymbol{W}_{\mathsf{MZ}} \in [0,\infty)^{M\times M}$

random $\longrightarrow$ subsampling

2

$\boldsymbol{X}$: randomly subsample n points with $n \sim m \log m$



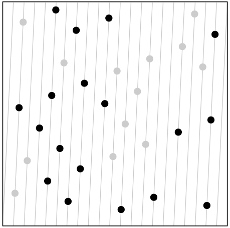
$\boldsymbol{X} = \{\boldsymbol{x}^i\}_{i\in J}$

$\boldsymbol{W} \in [0,\infty)^{|\boldsymbol{X}|\times|\boldsymbol{X}|}$

BSS $\longrightarrow$ subsampling

3

$\boldsymbol{X}'$: BSS subsample n' points with $n' \sim m$



$\boldsymbol{X}' = \{\boldsymbol{x}^i\}_{i\in J'}$

$\boldsymbol{W}' \in [0,\infty)^{|\boldsymbol{X}'|\times|\boldsymbol{X}'|}$

Central theorem for rank-1 lattices

Theorem for rank-1 lattices [B, Kämmerer, Potts, T. Ullrich '22]

Let
- $s > 1/2$,
- $I \subset I_{\mathsf{MZ}} \subset \mathbb{Z}^d$ hyperbolic cross frequency index sets, $|I| \geq 3$,
- $\boldsymbol{X}_{\mathsf{MZ}}$ reconstructing rank-1 lattice for I_{MZ} with M points,
- $b > 1 + \frac{1}{|I|}$.

Then we construct $\boldsymbol{X}' \subset \boldsymbol{X}_{\mathsf{MZ}}$ with $|\boldsymbol{X}'| = n \leq \lceil b|I| \rceil$ such that

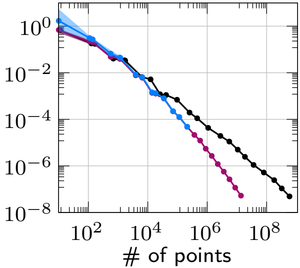
$$\sup_{\|f\|_{H^s_{\mathsf{mix}}} \leq 1} \|f - S_I^{\boldsymbol{X}'} f\|_{L_2} \leq C_{d,s,b}\Big(n^{-s}(\log n)^{(d-1)s+1/2} + M^{-s/2}(\log M)^{\frac{(d-2)s+d}{2}}\Big)$$

with probability $1 - 6/|I|$.

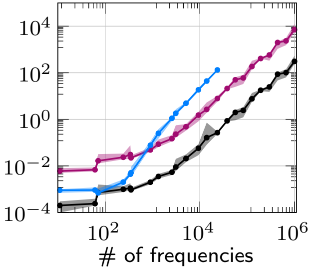
Numerical experiment for random subsampling

- $d = 5$, f tensorized bumb function in $H^s_{\text{mix}}(\mathbb{T}^5)$ for $s < 3/2$
- initial $\boldsymbol{X}_{\text{MZ}}$: fast probabilistic CBC rank-1 lattice construction [Kämmerer '20]

L_2-error



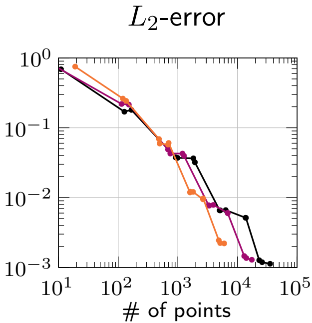
time in seconds

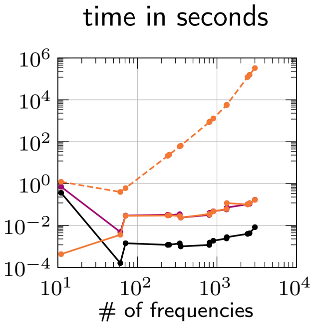


black: full rank-1 lattice
azure: continuously random
magenta: randomly subsampled

Numerical experiment for random + BSS subsampling

- $d = 5$, f tensorized bumb function in $H^s_{\text{mix}}(\mathbb{T}^5)$ for $s < 3/2$
- initial $\boldsymbol{X}_{\text{MZ}}$: fast probabilistic CBC rank-1 lattice construction [Kämmerer '20]





black: full rank-1 lattice
magenta: randomly subsampled
orange: BSS subsampled

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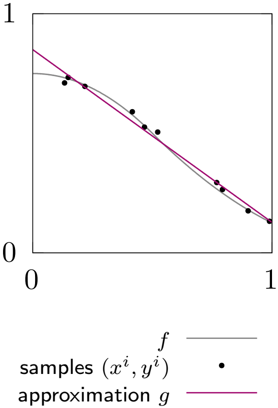
Learning setting

Given:

- $f\colon D \to \mathbb{C}$,
- points $\boldsymbol{x}^1, \dots, \boldsymbol{x}^n \in D$ distributed w.r.t. $\mathrm{d}\varrho_S$,
- samples $\boldsymbol{y} = (\underbrace{f(\boldsymbol{x}^1) + \varepsilon_1}_{y_1}, \dots, \underbrace{f(\boldsymbol{x}^n) + \varepsilon_n}_{y_n})^\mathsf{T}$.

Goal: find approximation $g\colon D \to \mathbb{C}$ with

- small L_2-error, i.e., $\int_D |f-g|^2 \,\mathrm{d}\varrho_T$ small.



Domain adaptation

- source measure $\mathrm{d}\varrho_S$ and target measure $\mathrm{d}\varrho_T$ may differ, e.g.
  - train a self learning car
  - train a spam filter
  - diagnostic models trained on earlier diseases predicting data associated with new viruses

Domain adaptation

- source measure $\mathrm{d}\varrho_S$ and target measure $\mathrm{d}\varrho_T$ may differ, e.g.
  - train a self learning car
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Radon-Nikodym theorem

Let
- ϱ_T and ϱ_S two σ-finite measures and
- ϱ_S be absolutely continuous w.r.t. ϱ_T.

Then there exists a measurable function $\beta\colon D \to [0, \infty)$, s.t.

$$\varrho_T(A) = \int_A \beta \,\mathrm{d}\varrho_S \quad \forall A \subset D\,.$$

Error bound in the learning setting

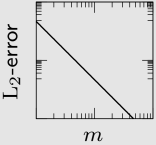
Theorem (Error bound without noise) [B 2022]

Let
- $f\colon D \to \mathbb{C}$, $x^1, \dots, x^n$, $n \in \mathbb{N}$ be points drawn according $\mathrm{d}\varrho_S$,
- $\beta = \varrho_T / \varrho_S$,
- $\boldsymbol{f} = (f(x^1), \dots, f(x^n))^\mathsf{T}$ function values,
- $t \ge 0$, $V = \operatorname{span}\{\eta_1, \dots, \eta_{m-1}\}$ satisfying

$$6 \Big\| \beta \sum_{k=1}^{m-1} |\eta_k|^2 \Big\|_\infty (\log(m) + t) \le n\,.$$

Then we have with probability exceeding $1 - 2\exp(-t)$:

$$\|f - S_{\boldsymbol{X}} \boldsymbol{f}\|_{L_2}^2 \le 8 \left(1 + \sqrt{\frac{t \| \sum_k |\eta_k|^2 \|_\infty}{n}} \right)^2 \|f - P(f, V)\|_{L_2}^2\,.$$



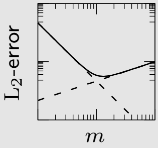
Error bound in the learning setting

Theorem (Error bound with noise) [B 2022]

Let
- the assumptions from the theorem before hold,
- $\boldsymbol{y} = (f(x^1) + \varepsilon_1, \dots, f(x^n) + \varepsilon_n)^\mathsf{T}$ noisy function values,
- $\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_n)^\mathsf{T}$ mean-zero rv's with $\mathbb{E}(|\varepsilon_i|^2) \le \sigma^2$ and $\|\boldsymbol{\varepsilon}\|_\infty \le B$.

Then we have with probability exceeding $1 - 3\exp(-t)$:

$$\|f - S_{\boldsymbol{X}}\boldsymbol{y}\|_{L_2}^2 \le 14\left(1 + \sqrt{\frac{t\|\sum_k |\eta_k|^2\|_\infty}{n}}\right)^2 \|f - P(f,V)\|_{L_2}^2 + 4\|\beta\|_\infty\left(\frac{m}{n}\left(14B\sqrt{t\sigma^2} + \sigma^2\right) + \frac{128B^2t}{n}\right).$$



Example on $[0, 1]$

	Chebyshev	**Legendre**	**H^1-basis**	**H^2-basis**
ϱ_T	$\frac{\mathrm{d}x}{\sqrt{1-(2x-1)^2}}$	$\mathrm{d}x$	$\mathrm{d}x$	$\mathrm{d}x$
ONB	$\cos(k \arccos(\cdot))$	$\frac{\mathrm{d}^k}{\mathrm{d}x^k}(x^2-1)^k$	$\cos(\pi k x)$	it's difficult
assumption	$f, \ldots, f^{(s-1)}$ absolute continuous, $f^{(s)}$ bounded variation		$f \in H^1$	$f \in H^2$
$\lVert f - P(f, V_m) \rVert_{L_2}$	$\mathcal{O}(m^{-(s+1/2)})$		$\mathcal{O}(m^{-1})$	$\mathcal{O}(m^{-2})$

Example on $[0,1]$

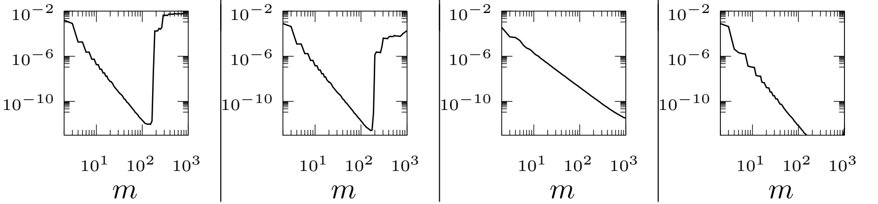
- given uniform points $\mathrm{d}\varrho_S = \mathrm{d}x$
- remember assumption $6\|\beta \sum_{k=1}^{m-1} |\eta_k|^2\|_\infty \log(m) \overset{!}{\leq} n$

	Chebyshev	Legendre	H^1-basis	H^2-basis
ϱ_T	$\frac{\mathrm{d}x}{\sqrt{1-(2x-1)^2}}$	$\mathrm{d}x$	$\mathrm{d}x$	$\mathrm{d}x$
ONB	$\cos(k \arccos(\cdot))$	$\frac{\mathrm{d}^k}{\mathrm{d}x^k}(x^2-1)^k$	$\cos(\pi k x)$	it's difficult
β	$\frac{1}{\sqrt{1-(2x-1)}}$	1	1	1
$\|\beta \sum_{k=1}^{m-1} \|\eta_k\|^2\|_\infty$	∞	m^2	$2m$	$6m$

Numerical experiment

- quadratic B-spline $f \in H^s$ for $s < 5/2$
- $x^1, \dots, x^n$ $10\,000$ uniformly distributed points
- no noise

Chebyshev	Legendre	H^1-basis	H^2-basis

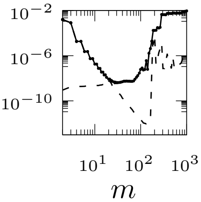


Example on $[0,1]$

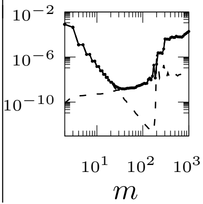
Numerical experiment

- quadratic B-spline $f \in H^s$ for $s < 5/2$
- $x^1, \ldots, x^n$ $10\,000$ uniformly distributed points
- 0.1% Gaussian noise in ε

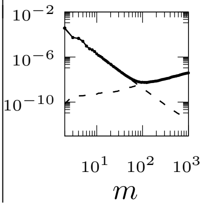
Chebyshev



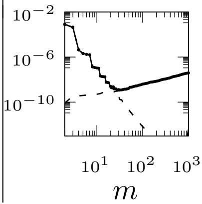
Legendre



H^1-basis



H^2-basis



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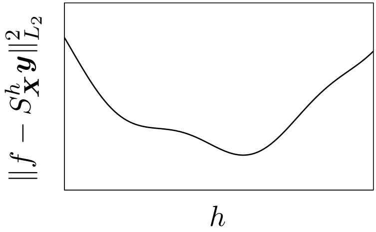
Setting

Given:

- samples $(x^i, y_i) \in D \times \mathbb{C}$ for $i = 1, \ldots, n$
- reconstruction method $S_{\boldsymbol{X}}$
- hypothesis space $H = \{V_h\}_{h \in H}$

Question 1: How big is the L_2-reconstruction error $\|f - S_{\boldsymbol{X}}\boldsymbol{y}\|_{L_2}^2$?

Question 2: Which reconstruction model $S_{\boldsymbol{X}}^h \boldsymbol{y}$ to choose?



Strategy

Purly data-driven method to approximate the risk

- validation set

train	test

Strategy

Purly data-driven method to approximate the risk

- validation set
- ”Probably the simplest and most widly used method for estimation prediction error is cross-validation.“ [Hastie 2001]

<table>
<tr><td>test</td><td colspan="4">train</td></tr>
<tr><td>train</td><td>test</td><td colspan="3">train</td></tr>
<tr><td colspan="2">train</td><td>test</td><td colspan="2">train</td></tr>
<tr><td colspan="3">train</td><td>test</td><td>train</td></tr>
<tr><td colspan="4">train</td><td>test</td></tr>
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Leave-one-out cross-validation [Stone 1974]

1. calculate the reconstructions $S^h_{\boldsymbol{X}_{-i}} \boldsymbol{y}_{-i}$ omitting the i-th sample
2. evaluate the residual of $S^h_{\boldsymbol{X}_{-i}} \boldsymbol{y}_{-i}$ in the i-th point $\boldsymbol{x}_i$

$$\left| (S^h_{\boldsymbol{X}_{-i}} \boldsymbol{y}_{-i})(\boldsymbol{x}_i) - y_i \right|$$

3. calculate the mean value with respect to all points

$$\mathrm{CV}(\boldsymbol{X}, \boldsymbol{y}, h) := \frac{1}{n} \sum_{i=1}^{n} \left| (S^h_{\boldsymbol{X}_{-i}} \boldsymbol{y}_{-i})(\boldsymbol{x}_i) - y_i \right|^2$$

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Fast computation

Theorem [Golub, Heath, Whaba 1979]

$$\mathrm{CV}(\boldsymbol{X}, \boldsymbol{y}, h) = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{[\boldsymbol{H}\boldsymbol{y} - \boldsymbol{y}]_i}{1 - h_{i,i}} \right|^2$$

with $h_{i,i}$ being the diagonal entries of $\boldsymbol{H} = \boldsymbol{L}(\boldsymbol{L}^* \boldsymbol{W} \boldsymbol{L})^{-1} \boldsymbol{L}^* \boldsymbol{W}$.

Theorem [Tasche, Weyrich 1996]

For $D = \mathbb{T}^d$, $\eta_{\boldsymbol{k}} = \exp(2\pi \mathrm{i} \langle \boldsymbol{k}, \boldsymbol{x} \rangle)$, and the full grid of frequencies and the full grid of points $\boldsymbol{X} = \{\boldsymbol{m}/N : \boldsymbol{m} \in \mathbb{Z}^d \cap \prod_{t=1}^{d} [0, N)\}$ the diagonal entries are given by

$$h_{i,i} = \frac{|I|}{N^d}\,.$$

Fast computation

Theorem (diagonal entries $h_{i,i}$) [B, Hielscher, Potts, 2019]

Let I be an frequency set and $\boldsymbol{X}$, $\boldsymbol{W}$ form an L_2-MZ inequality for $V = \operatorname{span}\{\exp(2\pi\mathrm{i}\langle\boldsymbol{k},\boldsymbol{x}\rangle)\}_{k\in I}$ with $A = B = 1$. Then

$$h_{i,i} = |I| w_i \,.$$

To compute the cross-validation score you need

- the least squares model evaluated at the given points $\boldsymbol{X} = \{\boldsymbol{x}^1, \dots, \boldsymbol{x}^n\}$
- the diagonal entries $h_{i,i}$

Fast computation

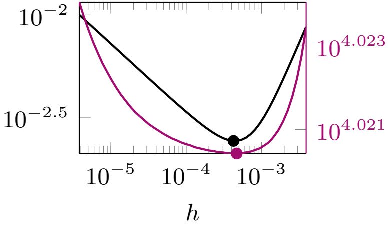
domain D	basis $\eta_{\boldsymbol{k}}$	diagonal elements $h_{i,i}$
$\mathbb{T}^d$	$\exp(2\pi\mathrm{i}\langle\boldsymbol{k},\boldsymbol{x}\rangle)$	$w_i\lvert I\rvert$
$[0,1]$	$\cos(k\arccos x)$	$\frac{w_i}{2}\left(\frac{2}{\pi}+2\sum_{k=1}^{K-1}\frac{\cos(2k\arccos x_i)+1}{\pi}\right)$
$\mathbb{S}^2$	$Y_{k,l}(\vartheta,\varphi)$	$\frac{w_i}{4\pi}m^2$
$\mathrm{SO}(3)$	$D_k^{ll'}(\boldsymbol{R})$	$\frac{w_i(m+1)(2m+1)(2m+3)}{24\pi^2}$

[B., Hielscher, Potts 2019]

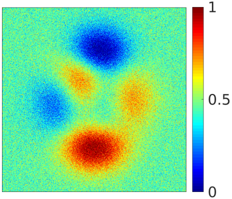
Numerical example

- 1024×1024 points
- $10\,\%$ Gaussian noise

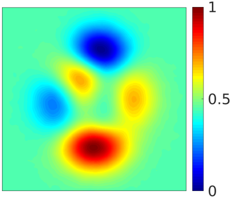
L_2-error $\|f - S^h_{\boldsymbol{X}}\boldsymbol{y}\|_{L_2}$ and
cross-validation score $\mathrm{CV}(\boldsymbol{X}, \boldsymbol{y}, h)$



noisy samples



reconstruction



Scattered points

given a good L_2-MZ inequality, we have

$$\boldsymbol{L}^*\boldsymbol{W}\boldsymbol{L} \approx \boldsymbol{I}\,,$$

which motivates

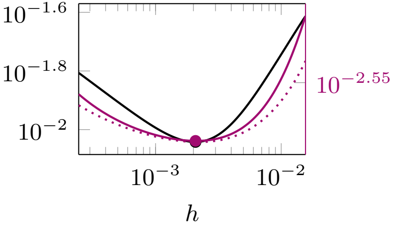
Definition

The approximated cross-validation score $\tilde{\mathrm{CV}}(\boldsymbol{X}, \boldsymbol{y}, h)$ is defined by

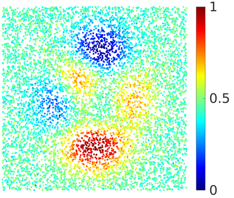
$$\tilde{\mathrm{CV}}(\boldsymbol{X}, \boldsymbol{y}, h) = \frac{1}{n}\sum_{i=1}^{n} \frac{[\boldsymbol{H}\boldsymbol{y} - \boldsymbol{y}]_i^2}{(1 - \tilde{h}_{i,i})^2} \quad \text{for} \quad \tilde{h}_{i,i} = w_i|I| \approx h_{i,i}.$$

- $8\,192$ uniformly scattered points
- $5\,\%$ Gaussian noise

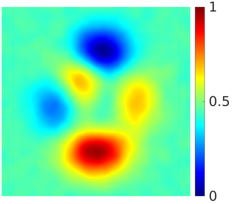
L_2-error $\|f - S^h_{\boldsymbol{X}}\boldsymbol{y}\|_{L_2}$ and
cross-validation score $\mathrm{CV}(\boldsymbol{X}, \boldsymbol{y}, h)$



noisy samples



reconstruction



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Problem

Given:

- samples $(x^i, y_i) \in D \times \mathbb{C}$ for $i = 1, \ldots, n$

Question: relation of

$$\|f - S_{\boldsymbol{X}}^h \boldsymbol{y}\|_{L_2}^2 = \int_D \left| S_{\boldsymbol{X}}^h \boldsymbol{y} - f \right|^2 \, \mathrm{d}\varrho_T$$

and

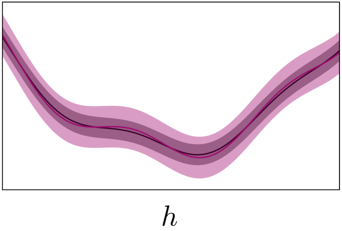
$$\mathrm{CV}(\boldsymbol{X}, \boldsymbol{y}, h) = \frac{1}{n} \sum_{i=1}^{n} \left| (S_{\boldsymbol{X}_{-i}} \boldsymbol{y}_{-i})(\boldsymbol{x}^i) - y_i \right|^2 .$$

Expectation of $\mathrm{CV}(\boldsymbol{X}, \boldsymbol{y}, h)$

Lemma

Let
- $\boldsymbol{y} = (f(x^1) + \varepsilon_1, \ldots, f(x^n) + \varepsilon_n)^\mathsf{T}$ noisy function values,
- $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_n)^\mathsf{T}$ mean-zero rv's with $\mathbb{E}(|\varepsilon_i|^2) = \sigma^2$.

$$\mathbb{E}_{\boldsymbol{X},\boldsymbol{y}}\Big(\mathrm{CV}(\boldsymbol{X}, \boldsymbol{y}, h)\Big) = \mathbb{E}_{\boldsymbol{X}_{-1},\boldsymbol{y}_{-1}}\Big(\|f - S_{\boldsymbol{X}_{-1}}\boldsymbol{y}_{-1}\|_{L_2}^2\Big) + \sigma^2 .$$



$\|f - S_{\boldsymbol{X}_{-1}}\boldsymbol{y}_{-1}\|_{L_2}^2 + \sigma^2$

$\mathrm{CV}(\boldsymbol{X}, \boldsymbol{y}, h)$

Theoretical foundation

Theorem

Let
- $\boldsymbol{x}^1, \ldots, \boldsymbol{x}^n$ be distributed w.r.t. ϱ_S,
- $|y_i| \leq M$, and

Then

$$\left| \mathbb{E}_{\boldsymbol{X}_{-1}}(\|f - S^h_{\boldsymbol{X}_{-1}} \boldsymbol{y}_{-1}\|^2_{L_2}) + \sigma^2 - \mathrm{CV}(\boldsymbol{X}, \boldsymbol{y}, h) \right| \leq 52M^2 \frac{n}{n - 4\| \sum_{k=1}^{m-1} |\eta_k|^2 \|_\infty} \frac{\log n \| \sum_{k=1}^{m-1} |\eta_k|^2 \|_\infty}{\sqrt{n}}$$

with probability exceeding $1 - 2/\sqrt{n}$.

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